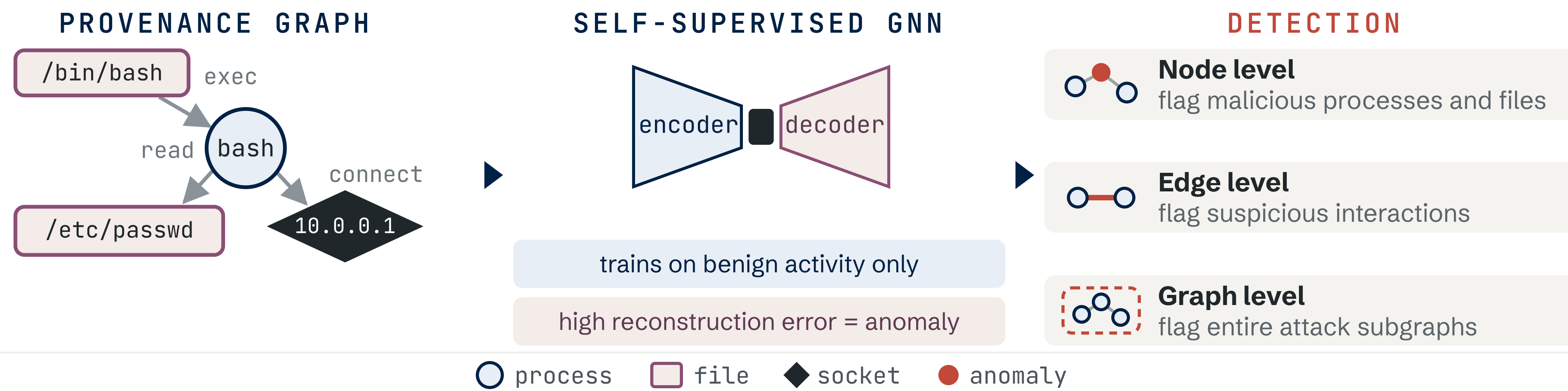


01 · BACKGROUND

Provenance-based Intrusion Detection Systems (PIDS)

A provenance graph links process, file and socket interactions causally, so a weeks-long attack shows up as one traceable path.
A PIDS flags unseen attacks as anomalies.



02 · THE PROBLEM

The PIDS research workflow is painful

Researchers rebuild everything, then evaluate under rules nobody shares.

Implement from scratch

Own training, inference and evaluation code, every single time.

Re-implement baselines

Reproduce other systems from papers with missing details and data.

Tune and ablate by hand

Hyperparameters and component variants, with no shared infrastructure.

Preprocess huge datasets

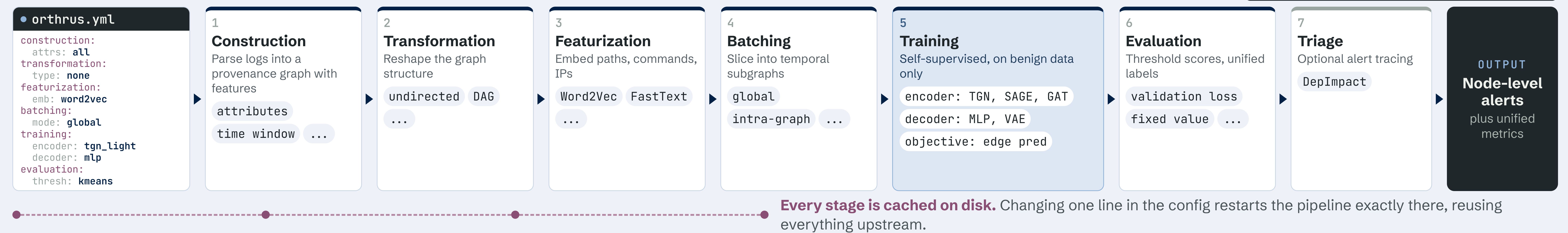
Provenance datasets typically need multi-TB storage and sever-level memory to preprocess.

Works rely on different ground truth labels. We need consistency for fair comparison.

03 · OUR SOLUTION

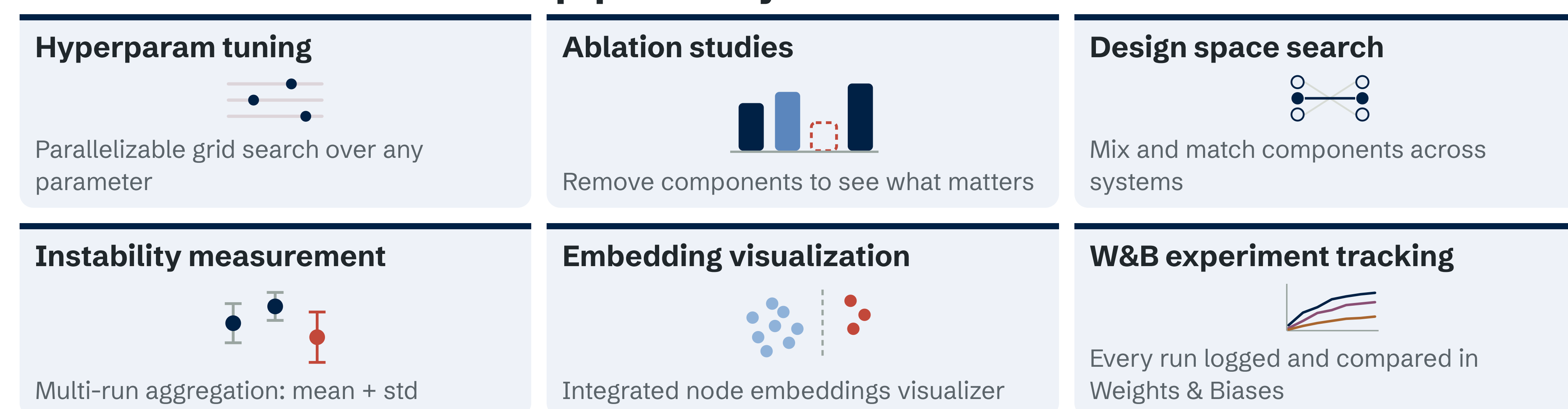
Accelerating PIDS research with a unified experimentation framework

One YAML file invokes any SOTA system for a fast, apples-to-apples benchmark, or swaps components to explore new system designs.



04 · FEATURES

Features built around the pipeline system



8 SOTA systems, one codebase

• ThreaTrace TIFS • Kairos S&P • NodLink NDSS • MAGIC USENIX Sec • Flash S&P • R-CAID S&P • Orthrusr USENIX Sec • VeLox USENIX Sec

15 datasets, preprocessed

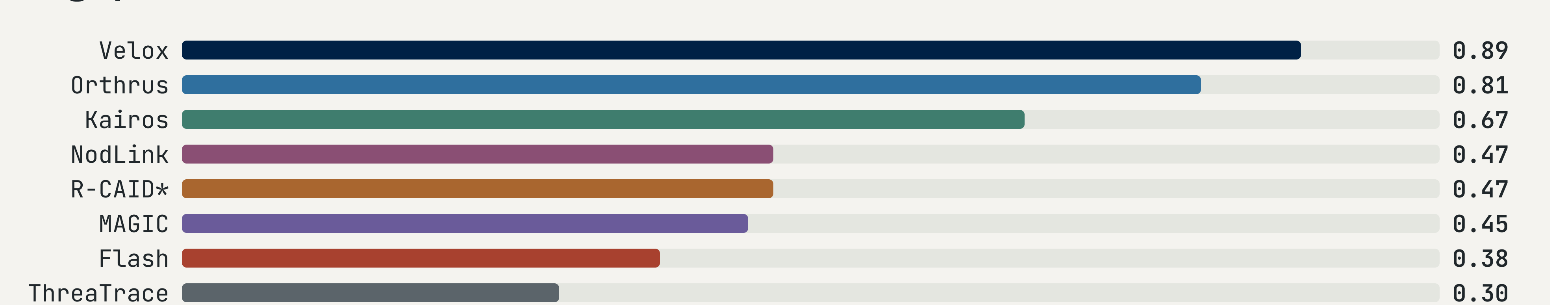
DARPA TC E3 and E5, OptC and more, all parsed once and shipped, so no one re-preprocesses hundreds of GB of logs.

05 · FINDINGS

What a fair benchmark reveals

detection at the node level (i.e., file, process, socket)

Average precision across 8 datasets



* R-CAID completed only 2 of 8 datasets within 24 h.

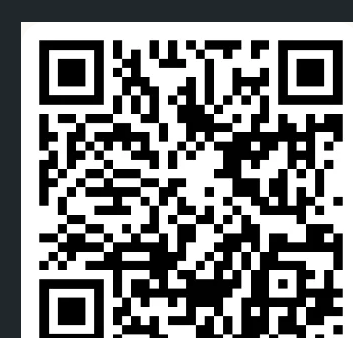
Three detection regimes

ThreaTrace, NodLink and MAGIC score **indiscriminately**. Kairos and Flash are **over-conservative**. VeLox, Orthrusr and R-CAID are **sparse but precise**.

Accuracy is not bought with cost

Memory spans **27 MB** (VeLox) to **29 GB** (Kairos), yet the heaviest systems detect the least.

8/8 datasets where VeLox catches at least one attack node, with very few false positives on most benchmarks.



The framework & datasets are open source!

docs [ubc-provenance.github.io/PIDSMaker](https://github.com/ubc-provenance/PIDSMaker)

repo github.com/ubc-provenance/PIDSMaker

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KDD 2026 (Datasets & Benchmarks Track)

Jeju, Korea